

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Change, Update]	178068	49.231127
1	[Git, Change, Create]	52544	14.527036
2	[Git, Change, Delete]	27895	7.712235
3	[Java, ByteCode, Member, Method]	13494	3.730737
4	[Java, ByteCode, Parameter]	13293	3.675165
5	[Git, Commit]	13289	3.674060
6	[File, Git]	11096	3.067753
7	[Git, Change, Rename]	10752	2.972646
8	[Java, ByteCode, Bound, ParameterizedType]	7279	2.012452
9	[Java, ByteCode, Bound]	7240	2.001670
10	[Java, ByteCode, Member, Field]	3609	0.997794
11	[Java, ByteCode, Bound, WildcardType]	2947	0.814768
12	[Java, Value, ByteCode, Annotation]	2931	0.810345
13	[Xml, Element]	2162	0.597736
14	[Java, ByteCode, Member, Method, Constructor]	2120	0.586124
15	[Xml, Text]	1450	0.400887
16	[Java, ByteCode, Bound, TypeVariable]	1111	0.307162
17	[Java, ByteCode, Member, Method, Lambda]	968	0.267627
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	888	0.245509
19	[Type, File, Java, ByteCode, Class]	846	0.233897
20	[Json, Key]	702	0.194085
21	[Value, Json, Scalar]	685	0.189385
22	[Java, Value, ByteCode, Primitive]	672	0.185790
23	[Type, File, Java, ByteCode, JavaType]	649	0.179431
24	[Java, ByteCode, Member, Method, GenericDeclar...	578	0.159802
25	[Type, File, Java, ByteCode, ExternalType]	411	0.113631
26	[Git, Change, Copy]	346	0.095660
27	[Author, Git, Person]	299	0.082666
28	[Value, Array]	287	0.079348
29	[Committer, Git, Person]	252	0.069671

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

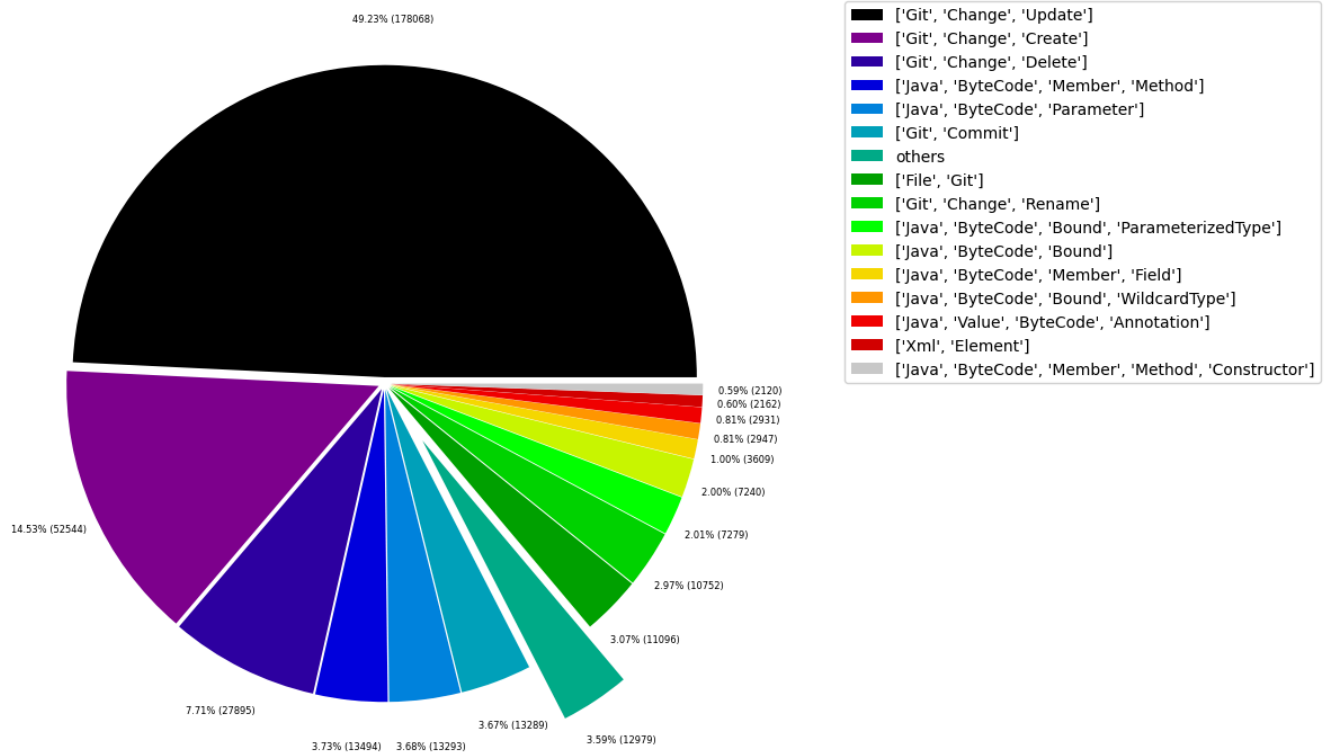


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000276
1	[Package, File, Json, NPM]	1	0.000276
2	[Repository, File, Git]	1	0.000276
3	[File, TS, Scan]	1	0.000276
4	[File, Json]	2	0.000553
5	[File]	3	0.000829
6	[Java, ByteCode, Member, Method, GenericDeclar...	4	0.001106
7	[Maven, Exclusion]	5	0.001382
8	[Value, Array, Json]	6	0.001659
9	[Dependency, NPM]	7	0.001935
10	[Type, File, Java, ByteCode, Void]	9	0.002488
11	[File, Maven, Xml, Pom, Document]	9	0.002488
12	[Java, ManifestSection]	9	0.002488
13	[File, Java, Manifest]	9	0.002488
14	[File, Artifact, Jar, Archive, Zip, Java]	9	0.002488
15	[File, Java, ServiceLoader]	10	0.002765
16	[File, Java, Properties]	12	0.003318
17	[Maven, ExecutionGoal]	16	0.004424
18	[Maven, PluginExecution]	16	0.004424
19	[Xml, Attribute]	18	0.004977
20	[jQAssistant, Rule, Concept]	19	0.005253
21	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.005806
22	[Maven, Configuration]	21	0.005806
23	[Maven, Plugin]	21	0.005806
24	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.006082
25	[Type, File, Java, ByteCode, Enum]	28	0.007741
26	[Type, File, Java, ByteCode, PrimitiveType]	30	0.008294
27	[Xml, Namespace]	36	0.009953
28	[Type, File, Java, ByteCode, Annotation]	44	0.012165
29	[Git, Branch]	47	0.012994

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

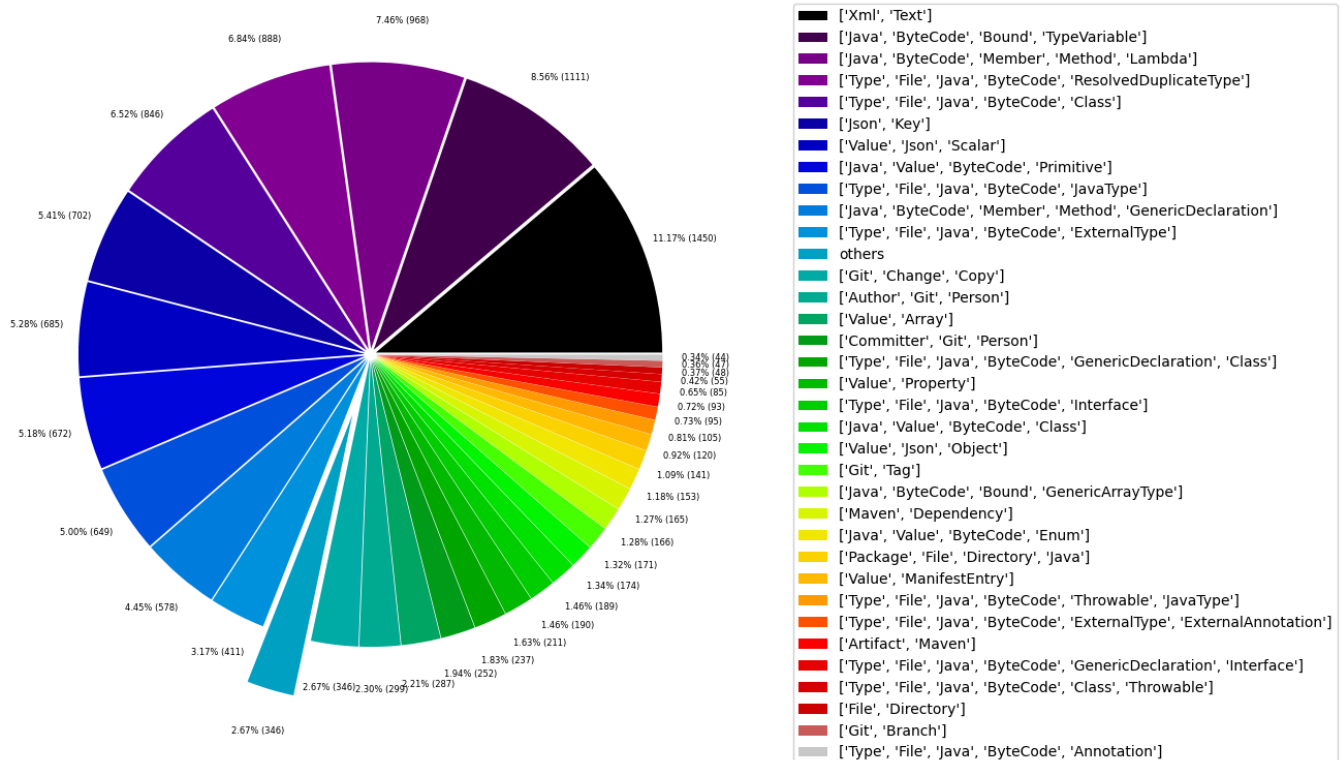


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

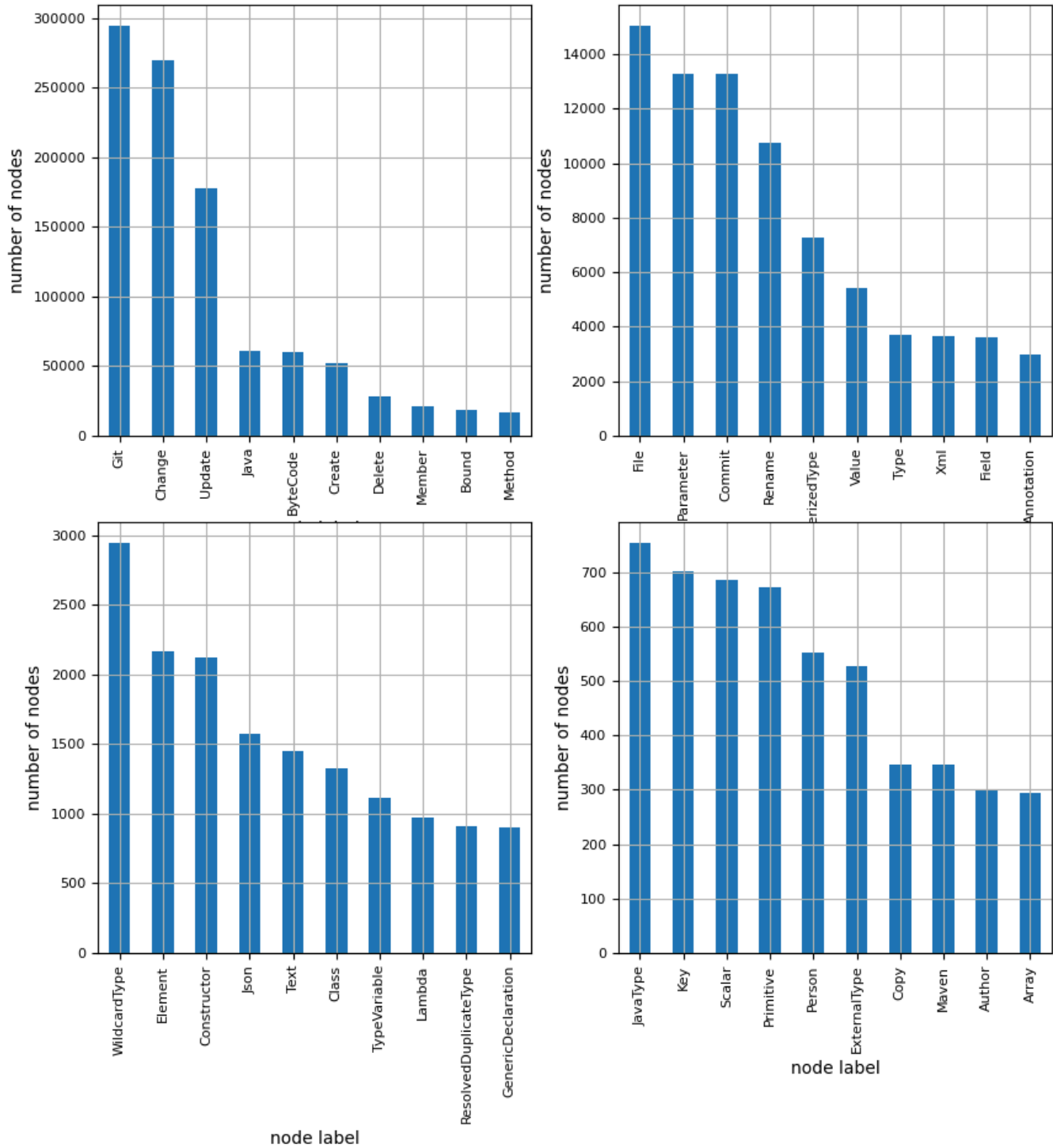
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	294760	81.493401
1	Change	269605	74.538704
2	Update	178068	49.231127
3	Java	60659	16.770621
4	ByteCode	60469	16.718091
5	Create	52544	14.527036
6	Delete	27895	7.712235
7	Member	20773	5.743189
8	Bound	18743	5.181947
9	Method	17164	4.745395
10	File	15057	4.162865
11	Parameter	13293	3.675165
12	Commit	13289	3.674060
13	Rename	10752	2.972646
14	ParameterizedType	7279	2.012452
15	Value	5428	1.500699
16	Type	3715	1.027100
17	Xml	3675	1.016041
18	Field	3609	0.997794
19	Annotation	2975	0.822509
20	WildcardType	2947	0.814768
21	Element	2162	0.597736
22	Constructor	2124	0.587230
23	Json	1570	0.434064
24	Text	1450	0.400887
25	Class	1327	0.366881
26	TypeVariable	1111	0.307162
27	Lambda	968	0.267627
28	ResolvedDuplicateType	910	0.251591
29	GenericDeclaration	904	0.249932
30	JavaType	754	0.208461
31	Key	702	0.194085
32	Scalar	685	0.189385
33	Primitive	672	0.185790
34	Person	551	0.152337
35	ExternalType	527	0.145702
36	Copy	346	0.095660
37	Maven	346	0.095660
38	Author	299	0.082666
39	Array	293	0.081007

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1122333

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	269605	24.021837
1	MODIFIES	269605	24.021837
2	UPDATES	178068	15.865879
3	CREATES	63642	5.670510
4	DELETES	38647	3.443452
5	INVOKES	36776	3.276746
6	COMMITTED	26578	2.368103
7	DEPENDS_ON	22492	2.004040
8	OF_TYPE	21920	1.953075
9	DECLARES	21234	1.891952
10	OF_RAW_TYPE	17353	1.546154
11	HAS_PARENT	16074	1.432195
12	HAS	14429	1.285626
13	HAS_COMMIT	13289	1.184051
14	RETURNS	12844	1.144402
15	HAS_FILE	11096	0.988655
16	RENAMES	10752	0.958004
17	READS	9378	0.835581
18	HAS_ACTUAL_TYPE_ARGUMENT	8407	0.749065
19	HAS_NEW_NAME	6310	0.562222
20	OF_GENERIC_TYPE	5996	0.534244
21	RESOLVES_TO	5270	0.469558
22	SIMILAR	4078	0.363350
23	WRITES	3944	0.351411
24	CONTAINS	3936	0.350698
25	RETURNS_GENERIC	3593	0.320137
26	ANNOTATED_BY	2919	0.260083
27	REQUIRES	2230	0.198693
28	HAS_FIRST_CHILD	2162	0.192634
29	HAS_LAST_CHILD	2162	0.192634

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

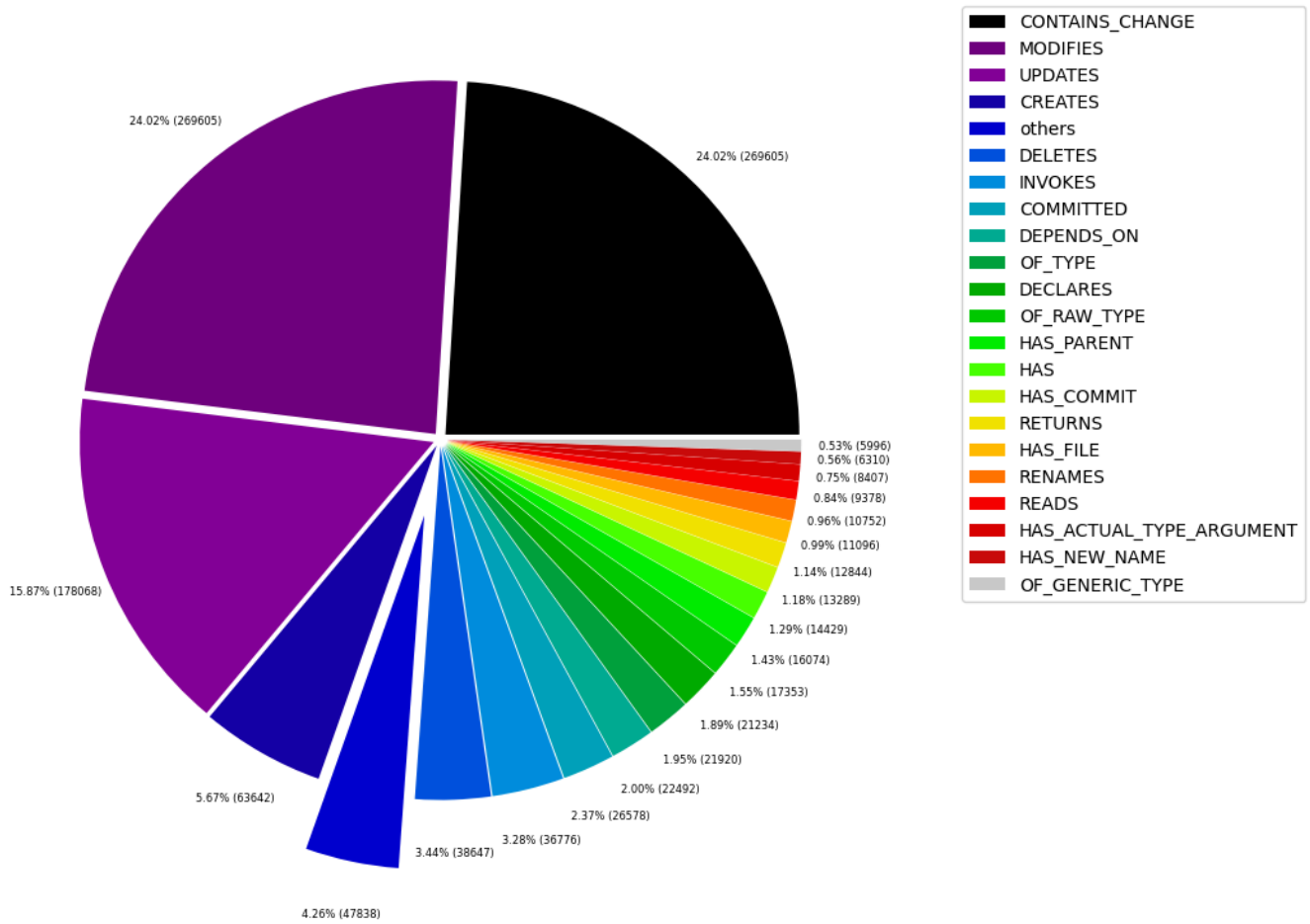


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

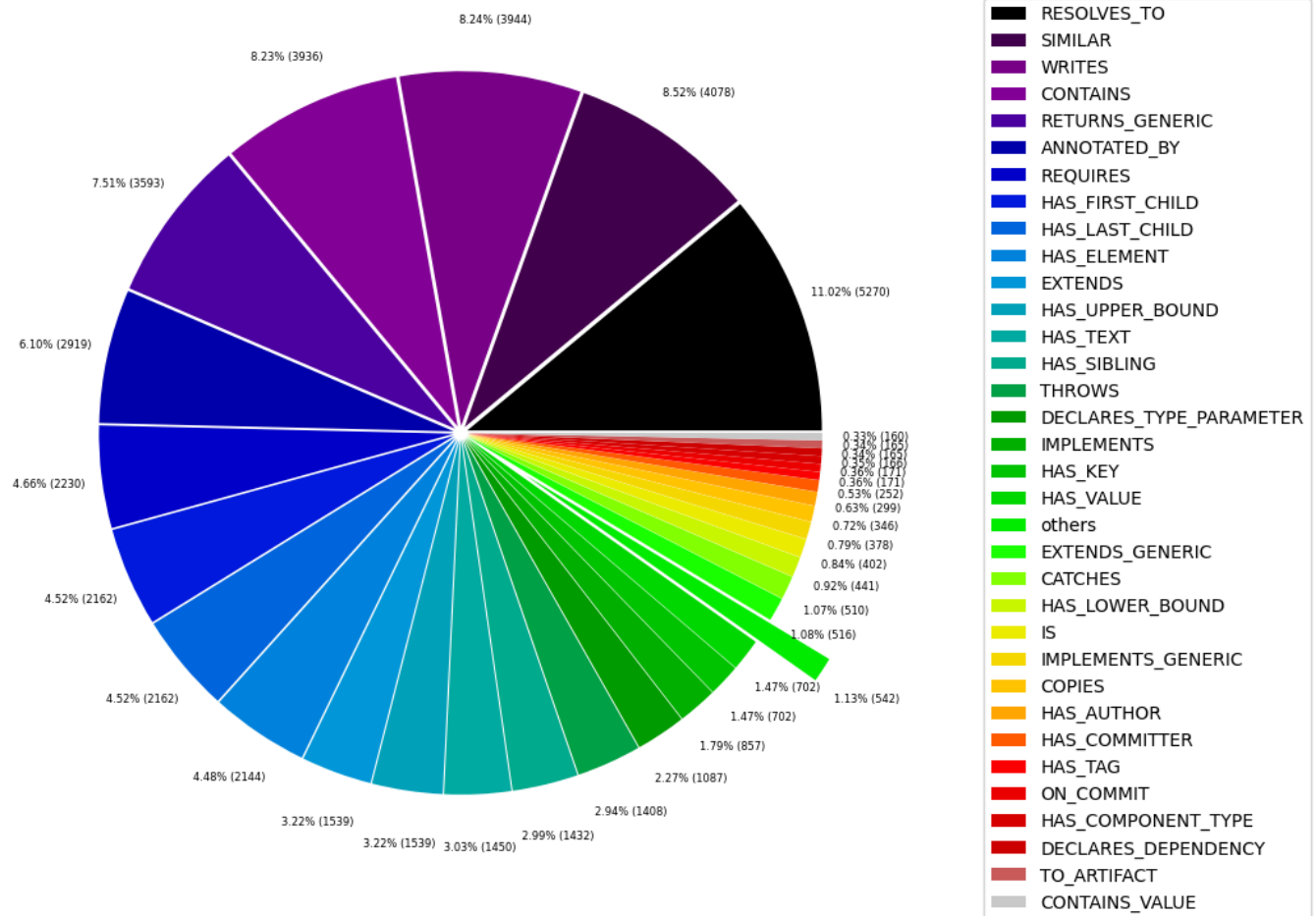
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000089
1	THROWS_GENERIC	5	0.000446
2	EXCLUDES	5	0.000446
3	DECLARES_DEV_DEPENDENCY	7	0.000624
4	DESCRIBES	9	0.000802
5	HAS_ROOT_ELEMENT	11	0.000980
6	HAS_GOAL	16	0.001426
7	HAS_EXECUTION	16	0.001426
8	OF_NAMESPACE	18	0.001604
9	HAS_ATTRIBUTE	18	0.001604
10	INCLUDES_CONCEPT	19	0.001693
11	IS_ARTIFACT	21	0.001871
12	HAS_CONFIGURATION	21	0.001871
13	USES_PLUGIN	21	0.001871
14	REQUIRES_TYPE_PARAMETER	24	0.002138
15	REQUIRES_CONCEPT	28	0.002495
16	DECLARES_NAMESPACE	36	0.003208
17	HAS_DEFAULT	39	0.003475
18	HAS_BRANCH	47	0.004188
19	HAS_HEAD	48	0.004277
20	COPY_OF	132	0.011761
21	CONTAINS_VALUE	160	0.014256
22	DECLARES_DEPENDENCY	165	0.014702
23	TO_ARTIFACT	165	0.014702
24	HAS_COMPONENT_TYPE	166	0.014791
25	ON_COMMIT	171	0.015236
26	HAS_TAG	171	0.015236
27	HAS_COMMITTER	252	0.022453
28	HAS_AUTHOR	299	0.026641
29	COPIES	346	0.030829

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Change, Update]	MODIFIES	[File, Git]	178068	178068	
1	[Git, Change, Update]	UPDATES	[File, Git]	178068	178068	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Update]	178068	13289	
3	[Git, Change, Create]	MODIFIES	[File, Git]	52544	52544	
4	[Git, Change, Create]	CREATES	[File, Git]	52544	52544	
5	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	52544	13289	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	27895	13289	
7	[Git, Change, Delete]	MODIFIES	[File, Git]	27895	27895	
8	[Git, Change, Delete]	DELETES	[File, Git]	27895	27895	
9	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	22590	13404	
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	16065	13289	
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	13289	1	
12	[Author, Git, Person]	COMMITTED	[Git, Commit]	13289	299	
13	[Committer, Git, Person]	COMMITTED	[Git, Commit]	13289	252	
14	[Repository, File, Git]	HAS_FILE	[File, Git]	11096	1	
15	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	10752	13289	
16	[Git, Change, Rename]	CREATES	[File, Git]	10752	10752	
17	[Git, Change, Rename]	DELETES	[File, Git]	10752	10752	
18	[Git, Change, Rename]	MODIFIES	[File, Git]	10752	10752	
19	[Git, Change, Rename]	RENAMES	[File, Git]	10752	10752	
20	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	8517	13404	
21	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	8446	13404	
22	[File, Git]	HAS_NEW_NAME	[File, Git]	6310	11096	
23	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6156	13293	
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3465	7240	
25	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3110	7279	
26	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2947	7279	
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, Bound, ParameterizedType]	2695	13293	
28	[Java, ByteCode, Member, Method, Constructor]	WRITES	[Java, ByteCode, Member, Field]	2519	2120	
29	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, TypeVariable]	2474	7279	

Graph Density

```
total_number_of_nodes (vertices): 361698
total_number_of_relationships (edges): 1122333
-> total directed graph density: 8.578882482473568e-06
-> total directed graph density in percent: 0.0008578882482473567
```